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ScienceDirect

Procedia Computer Science 277 (2026) 1019–1030

Procedia Computer Science

www.elsevier.com/locate/procedia

7th International Conference on Industry of the Future and Smart Manufacturing
(former International Conference on Industry 4.0 and Smart Manufacturing)

Explainable and Scalable Job Recommendation System using Transformer-based Text Summarization, Cross-Encoder Semantic Similarity, and Ensemble Learning

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Abstract

This research investigates the impact of combining BERT-based semantic similarity with ensemble classifiers and explainable AI (XAI) on improving recommendation accuracy and interpretability in job matching systems aligned with the Industry 4.0 paradigm. We propose a hybrid system integrating BERT-based summarization, Cross-Encoder semantic scoring, and ensemble classification (XGBoost, Random Forest, AdaBoost) to deliver accurate and transparent recommendations. Leveraging a large LinkedIn dataset (1.3M job postings), the system captures joint semantic relevance and structural features such as job seniority and skill density. Experimental results show that XGBoost achieves 93.4% accuracy with an RMSE of 0.237. Explainability is achieved through SHAP and LIME, ensuring accountability in AI-assisted hiring. The framework also takes into account the recommendation of emerging roles such as robotics engineers, IoT analysts, and digital twin specialists; critical to smart manufacturing and future industry needs. Furthermore, the proposed system addresses semantic and transparency gaps in large-scale, real-world recruitment scenarios.

Keywords: Artificial intelligence, Recommender systems, BERT Cross-Encoder, Explainable AI (XAI), Semantic Similarity

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1877-0509

1. Introduction

In alignment with the Industry 4.0 paradigm, there is a growing need for intelligent recruitment systems capable of identifying talent for emerging roles such as robotics engineers, Internet of Things (IoT) analysts, and digital twin specialists. These positions demand highly specialized skills and contextual understanding that traditional recommender systems struggle to capture, highlighting the urgency for semantically rich and explainable AI solutions.

Traditional job recommender systems have long relied on keyword-based retrieval techniques such as Term Frequency - Inverse Document Frequency (TF-IDF) and Cosine similarity, which suffer from surface-level lexical matching and an inability to capture contextual semantics [1], [2]. These limitations often result in inaccurate job-user matching, especially when synonyms, contextual cues, or domain-specific jargon are involved [3], [4]. Furthermore, many existing systems lack transparency, making them susceptible to algorithmic bias and limiting user trust and system accountability [5], [6]. The growing demand for fair and accountable Artificial Intelligence (AI) in recruitment has underscored the importance of incorporating Explainable AI (XAI) in recommender systems [7], [8], [9]. Techniques such as SHapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) have proven effective in surfacing model logic in post-hoc analysis, making AI decisions more interpretable for non-technical stakeholders [10], [11].

Semantic similarity is a cornerstone in enhancing recommendation accuracy. However, traditional similarity metrics like Cosine similarity and TF-IDF do not account for deeper semantic relationships or word order, which reduces their efficacy in complex, real-world recommendation scenarios [4], [12]. Recent studies have proposed approaches that utilize transformer-based models, especially Bidirectional Encoder Representations from Transformers (BERT) variants, to overcome such deficiencies [13], [14].

Our previous research [15] demonstrated the utility of BERT-based text summarization in enhancing job recommendation quality by reducing textual noise and emphasizing contextually relevant content. This model processes input pairs jointly and learns complex relationships, outperforming Cosine similarity and even BERTScore in re-ranking and retrieval contexts [16], [17]. However, a significant research gap remains in the integration of transformer-based summarization with explainable and scalable job recommendation frameworks. Most prior systems overlook the benefits of combining semantic understanding, structured data features, and explainability within a unified architecture. Furthermore, few studies address real-world scalability using large industrial datasets like the 1.3M LinkedIn job postings used in this study. To fill this gap, we pose the following research question: “Can combining BERT-based semantic similarity with ensemble classifiers and XAI improve recommendation accuracy and interpretability over traditional methods?”

In this study, we build upon our previous findings by adopting a Cross-Encoder BERT architecture to compute semantic relevance between users and jobs through pairwise interaction modeling. The proposed model has been enriched with structured features such as job seniority and skill count and employ ensemble learning for classification. Finally, several explainability modules; namely SHAP, LIME, and ELI5, have been incorporated to offer interpretable recommendations that can support fair and transparent decision-making in hiring.

The remainder of this paper is structured as follows. Section 2 outlines the Related Work. Section 3 presents the methodology, encompassing the dataset, BERT-based summarization, semantic scoring via Cross-Encoder, ensemble learning models, and explainability frameworks. Section 4 discusses the experimental results, including performance metrics, Top-K recommendations, and interpretability analysis. Section 5 concludes the paper by summarizing contributions, identifying limitations, and proposing future enhancements.

2. Related Work

Job recommendation systems have evolved substantially over the past few years, shifting from basic content-based filtering to deep learning and Large Language Models (LLMs). The researchers in early studies, such as [21] and [22] focused on rule-based systems, TF-IDF matching, and collaborative filtering with moderate performance in real-world settings. Later efforts adopted machine learning and neural approaches to improve accuracy, such as CNN-based pipelines [16], Bi-LSTM with attention mechanisms [13][21], and knowledge graph-enhanced models [15].

Recent works have explored leveraging transformers and LLMs. Ghosh et al. [11] compared deterministic, LLM-guided, and hybrid models to better capture soft skills and contextual information. Du et al. [9] used GANs to enhance resume quality but lacked interpretability. Other researchers such as Yingpeng Du [9] and Patil et al. [10] attempted to improve performance via LLMs or ensemble models, but scalability and transparency remained key limitations.

Several studies, such as [12],[14],[17], and [18], integrated user metadata or attention mechanisms for personalization. However, most of these studies failed to combine semantic summarization, structured feature engineering, and XAI modules in a single architecture. Our previous work [15] partially addressed this challenge by applying BERT-based summarization and Cosine similarity, however, it did not implement deep pairwise modeling, classification, or explainability.

In this study, we advance the previously proposed work by:

- Employing Cross-Encoder BERT for semantic pairwise relevance scoring,
- Fusing structured features (skills, seniority) with BERT embeddings,
- Integrating ensemble classifiers (XGBoost, RF, AdaBoost) for classification,
- Applying SHAP, LIME, and ELI5 XAI techniques to improve interpretability for decision-makers.

This combined approach aims to close gaps in semantic depth, scalability, and explainability; essential for recruitment systems targeting specialized Industry 4.0 professions. A comparative analysis highlighting research gaps and limitations in existing research is summarized in Table 1.

Table 1. Job Recommendation Systems Comparative Analysis

Study	Technique & Dataset	Key Contribution	Limitations
[24] (2022)	Bidirectional matching with ML	Recruiter-candidate reciprocity modeling	No personalization
[10] (2023)	Hybrid + LLMs	Contextual fit & soft skill modeling	Poor scalability
[23] (2023)	Survey of RecSys techniques	Broad analysis of ML/DL methods	Lacks performance/implementation depth
[25] (2024)	GANs + LLM on real resumes	Resume refinement for low-quality inputs	No model transparency
[15] (2025)	BERT summarization + Cosine Similarity	Semantic condensation of job descriptions	No classification or XAI integration
Proposed Framework	Cross-Encoder + Ensemble + XAI + LinkedIn (1.3M)	Semantic scoring + feature fusion + transparency	Scalable & interpretable recommendation

3. Methodology

3.1. Dataset Description

The proposed system uses the publicly available 1.3M LinkedIn Jobs and Skills dataset from Kaggle [26], scraped in 2024. It contains metadata including job titles, descriptions, skills, seniority level, and location. After preprocessing, numerical features such as “TitleLength”, “SkillCount”, “IsSenior”, and “DescLength” were engineered to supplement semantic signals.

Despite providing an extensive foundation for scalable job recommendation, the utilized LinkedIn Jobs and Skills dataset is subject to certain constraints considering its inherent limitations and biases. The dataset is predominantly US-centric, reflecting the job market structure, role naming conventions, and language use in North America. Consequently, recommendations may be biased towards Western job taxonomies, limiting generalizability across other geographic or cultural contexts. Additionally, representation gaps exist across gender and cultural dimensions. This imbalance may skew relevance scores and job rankings, especially in global applications or equity-focused recruitment platforms. Moreover, despite the dataset’s inclusion of cutting-edge roles such as robotics engineers and digital twin specialists, which are aligned with the Industry 4.0 paradigm, it lacks representation for several emerging professions such as IoT analysts and prompt engineers. The absence of such roles may hinder the system’s ability to fully capture the breadth of next-generation employment needs. Future work should consider augmenting the dataset with more regionally diverse sources and incorporating synthetic or crowdsourced data for underrepresented roles to improve fairness and scope.

3.2. BERT-based Summarization

To reduce noise and extract meaningful content, the bert-extractive-summarizer was applied to job descriptions. This ensures that the semantic similarity computations focus on core content. BERT-based summarization captures syntactic and contextual nuances often missed by basic truncation or keyword extraction [16]. Prior research [21] has demonstrated that summarization improves embedding quality, ultimately boosting downstream recommendation accuracy.

3.3. Semantic Relevance using Cross-Encoder

We utilize cross-encoder/ms-marco-MiniLM-L-6-v2, a transformer model that processes user queries like "data scientist" or "data engineer" and job summaries as input pairs. Unlike Cosine similarity, the proposed model captures deep contextual interactions between input sequences. Cross-Encoder jointly encodes the job description and user query, enabling it to capture rich interactions between them [17]. This technique has demonstrated superior performance in ranking tasks across domains, including web search and conversational agents [18].

3.4. Feature Engineering

Feature engineering is the process of extracting, transforming, and selecting relevant variables from raw data to improve the performance of machine learning models. In the context of recommendation systems, feature engineering plays a vital role in capturing hidden patterns and enhancing the model's ability to differentiate between relevant and non-relevant items. In this context, engineered numerical features enrich the representation of each job posting, as follows:

- TitleLength: token count of job titles.
- SkillCount: number of listed skills.
- IsSenior: binary indicator for roles with "senior", "lead", or "manager".
- DescLength: character count of summarized descriptions.

These are combined with Cross-Encoder scores to formulate a unified input for classification models.

3.5. Ensemble Classifiers

Ensemble methods, such as Random Forest, Adaptive Boosting (AdaBoost), eXtreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), Categorical Boosting (CatBoost), Gradient Boosting Machine (GBM), provide powerful tools for building more reliable and accurate models [19]. The proposed system integrates three classifiers:

- **XGBoost**: captures complex feature interactions and provides native SHAP support. This gradient boosting framework is highlighted for its predictive power and integration with collaborative filtering methods, enhancing recommendation quality when combined with K-Nearest Neighbors (KNN) and Singular Value Decomposition (SVD) classifiers [20].
- **Random Forest**: aggregates decisions via bootstrapped trees. It is a widely used ensemble method, in the context of its performance compared to other ensemble techniques, including BoostForest, which outperformed it on various datasets [21].
- **AdaBoost**: emphasizes misclassified instances to improve robustness. This adaptive boosting technique is noted for its ability to improve model accuracy through iterative reweighting of training instances, contributing to better recommendations [22].

Each model outputs a probability of job relevance, which is aggregated with the CrossEncoder score using a weighted formula to compute the final job ranking.

3.6. Decision Explainability

To enhance interpretability of the employed classifiers decision, the proposed system integrated three XAI techniques::

- **SHapley Additive Explanations (SHAP)** is used for global feature attribution in XGBoost. It quantifies feature impact at the individual prediction level [10].
- **Local Interpretable Model-Agnostic Explanations (LIME):** provides local instance-level explanations for all classifiers [11].
- **ELI5 (Explain Like I'm 5) Framework:** aims to make complex AI systems more understandable through simplifying explanations for non-experts [24][25]. ELI5 provides a unified framework for inspecting model weights/feature importance across various classification models.

3.7. Proposed Transformer-based Ensemble Model

The proposed model, illustrated in Figure 1; follows a structured multi-stage pipeline designed to deliver context-aware and explainable job recommendations.

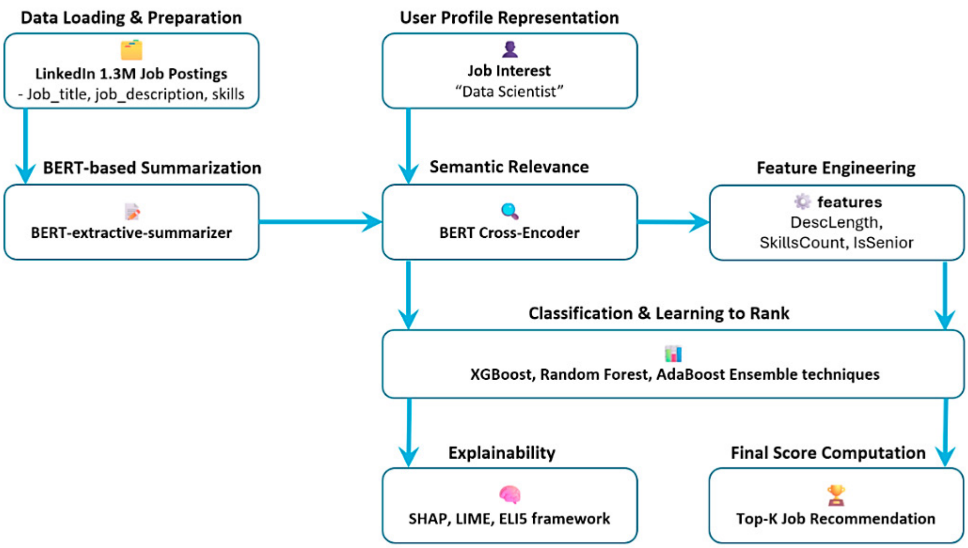


Fig. 1. Proposed job recommendation system

It begins with data loading and preprocessing, leveraging the large-scale LinkedIn dataset comprising 1.3M job postings enriched with metadata such as job titles, descriptions, and skills. In the second stage, BERT-based summarization is applied using the bert-extractive-summarizer to condense verbose job descriptions into semantically dense summaries, thereby enhancing downstream embedding quality. The third stage involves user profile representation, where the user's job interest (e.g., "data scientist") acts as a reference query for semantic scoring.

Subsequently, semantic relevance is computed using the Cross-Encoder model (cross-encoder/ms-marco-MiniLM-L-6-v2), which processes each user-job pair jointly to produce a context-rich similarity score that outperforms traditional metrics such as Cosine similarity and BERTScore. This is followed by a feature engineering stage, where numerical signals such as "DescLength", "TitleLength", "SkillCount", and "IsSenior" are extracted to represent structural characteristics of each job.

The next phase involves classification and learning to rank, where three ensemble models: XGBoost, Random Forest, and AdaBoost; are trained using both semantic and engineered features. Each model outputs a probability of relevance, and these scores are aggregated with the Cross-Encoder similarity to compute a final hybrid score, reflecting both semantic alignment and feature-based matching. The top-K job recommendations are selected by ranking postings based on this composite score.

Finally, the model includes an explainability layer comprising SHAP for global interpretability in tree-based models, LIME for local instance-level explanation across all classifiers, and ELI5 to provide simplified global feature importance.

Together, these stages ensure that the system not only generates accurate and personalized recommendations but also maintains transparency and interpretability, in addition to addressing key challenges in AI-driven recruitment systems.

3.8. System Scalability and Computational Complexity

To ensure the proposed recommendation system is scalable for real-world recruitment scenarios, we carefully optimized model selection, inference efficiency, and classifier training. Given the extended size of the LinkedIn dataset, balancing semantic performance with latency was crucial for production readiness.

For semantic similarity feature selection, MiniLM Cross-Encoder has been adopted based on its favorable trade-off between speed and performance. Unlike larger BERT-based Cross-Encoders, MiniLM delivers comparable semantic relevance while offering significantly lower computational overhead, making it suitable for large-scale inference. Table 2, shows the different Cross-Encoder models with their corresponding characteristics..

Table 2. Cross-Encoder model characteristics

Model	**Parameters**	**Inference Time (ms per pair)**	**Average Accuracy**	**Notes**
BERT-based Cross-Encoder	~110M	~135 ms	High	High accuracy, slow inference
DistilBERT Cross-Encoder	~66M	~88 ms	Medium-High	Faster, with mild accuracy drop
MiniLM-L-6-v2 (Chosen Model)	**33M**	**36 ms**	**High**	**Fastest, retains strong accuracy**

This efficiency allows processing numerous of job-user pairs with minimal delay, critical for near real-time recommendation systems.

Classifier Training on Subsamples

Given the memory and time complexity of ensemble classifiers (XGBoost and Random Forest), stratified subsampling method has been adopted for the proposed system to preserve class balance while reducing training load. Models were trained on representative subsets of 50k–100k labeled instances, which preserved performance (≤0.5% accuracy drop) while reducing training time by over 60%.

Inference Pipeline Optimization

The entire pipeline was optimized for batch processing, leveraging GPU acceleration where applicable. Jobs were semantically matched in batches of 128 using the Cross-Encoder, and classification was performed on the top-N candidates using preloaded ensemble models.

4. Results and Discussion

This section evaluates the performance and interpretability of the proposed explainable job recommendation system, which combines BERT-based semantic understanding with ensemble classifiers and post-hoc explainability techniques.

4.1. Performance evaluation

Table 3 presents the evaluation results for the three ensemble classifiers: XGBoost, Random Forest, and AdaBoost; based on Accuracy, Precision, Recall, F1 Score, and Root Mean Square Error(RMSE) performance metrics. XGBoost outperforms other implemented models with Accuracy (93.4%) and F1 Score (0.33), while AdaBoost showed the highest Precision (0.86) but suffered from extremely low Recall (0.06), indicating a trade-off between Precision and Recall. Random Forest offered a balanced trade-off between metrics but did not outperform XGBoost in any individual measure. The RMSE values further support this ranking, with XGBoost and Random Forest achieving lower errors (0.238 and 0.236, respectively) compared to AdaBoost (0.321). These results validate the selection of XGBoost as the most effective base learner in the ensemble.

Table 3. Performance measures of ensemble models

#	Model	Accuracy	Precision	Recall	F1 Score	RMSE
0	XGBoost	0.934	0.822	0.203	0.326	0.237
1	RandomForest	0.927	0.537	0.442	0.485	0.235
2	AdaBoost	0.925	0.855	0.056	0.106	0.321

4.2. Top-K Job Recommendations

Table 4 provides the Top 20 job recommendations for a user with a target profile of "data scientist" profession. Each recommendation includes the semantic similarity score (Cross-Encoder), classifier probabilities, and a final ensemble score. High Cross-Encoder scores (>8.3) confirm semantic relevance to the query, while classifier probabilities reflect how well structured features align with job relevance patterns. The job entitled "senior data warehousing specialist" achieved the highest final score (6.92), indicating consensus across both semantic and feature-based predictors. Notably, positions in fraud detection, NLP, and computational biology were also ranked highly, showcasing the model's ability to surface specialized roles that match inferred user intent.

Table 4. Job recommendation results for "data scientist" sample jobs

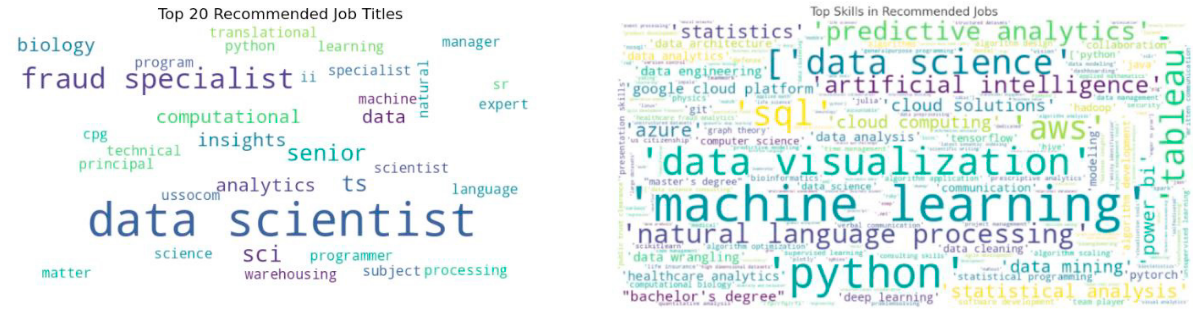
#	JobTitle	cross_score	xgb_score	rf_score	ada_score	final_score
1	data scientist: senior data warehousing specialist	9.127656	0.982134	1.00	0.732439	6.921
2	data scientist (computational biology)	8.808964	0.923509	0.99	0.748889	6.736
3	data scientist/senior data scientist, insights and analytics	8.742381	0.980664	1.00	0.732439	6.728
4	data scientist/senior data scientist, insights and analytics	8.742381	0.960056	0.99	0.732439	6.712
5	data scientist i/ii computational biology / machine learning	8.779306	0.852051	1.00	0.748889	6.690
6	data scientist/ sr. data scientist - (cpg)	8.711588	0.916047	0.99	0.748889	6.683
7	data scientist, translational data science (principal scientist)	8.699034	0.891680	1.00	0.748889	6.670
8	data scientist - fraud specialist	8.537883	0.927109	1.00	0.748889	6.607
9	data scientist - fraud specialist	8.537883	0.922938	1.00	0.748889	6.605
10	data scientist - fraud specialist	8.537883	0.922573	1.00	0.748889	6.605
11	data scientist - natural language processing	8.544134	0.912892	1.00	0.748889	6.603
12	data scientist - fraud specialist	8.537883	0.911195	1.00	0.748889	6.599
13	data scientist - fraud specialist	8.537883	0.910990	1.00	0.748889	6.599
14	data scientist - technical program manager	8.434526	0.979828	1.00	0.732439	6.573
15	data scientist (ts/sci)	8.387798	0.920054	1.00	0.717407	6.513
16	data scientist (ts/sci)	8.387798	0.911850	1.00	0.717407	6.509
17	data scientist (ts/sci)	8.387798	0.902241	1.00	0.717407	6.504
18	data scientist, subject matter expert	8.341642	0.892883	0.98	0.748889	6.482

19	data scientist (python programmer)	8.277657	0.933007	1.00	0.748889	6.480
20	data scientist (ussocom)	8.331856	0.891270	1.00	0.717407	6.470

4.3. Visualization of Recommendations and Skills

A visualizations of top recommended jobs and the corresponding skills are illustrated in the word clouds shown in Figure 2. It shows strong representation of terms like “*data scientist”*, “*machine learning”*, and “*senior”*, reinforcing the alignment between the model’s output and the user’s target role.

Moreover, further validation of the proposed system's effectiveness through a skill frequency analysis is depicted in Figure 2 (b). Important skills include “*Python”*, “*machine learning”*, “*SQL”*, and “*data analysis”*, which are directly aligned with the competencies expected in "data scientist" roles. This highlights the success of BERT-based summarization and cross-encoding in surfacing semantically meaningful job descriptions.



(a) Top 20 recommended job titles word cloud

(b) Top skills in recommended jobs

Fig. 2. Visualizations of recommended jobs and skills

4.4. Explainability Analysis

4.4.1. SHAP Analysis

SHAP was initially selected as a post-hoc explainability method due to its strong theoretical foundation in cooperative game theory and its ability to assign consistent and locally accurate attributions to model predictions.

However, applying SHAP to Random Forest and AdaBoost introduced technical challenges due to inconsistent support across different library versions. These limitations impacted the stability and interpretability of SHAP results for those models, reaffirming SHAP’s strongest compatibility with XGBoost.

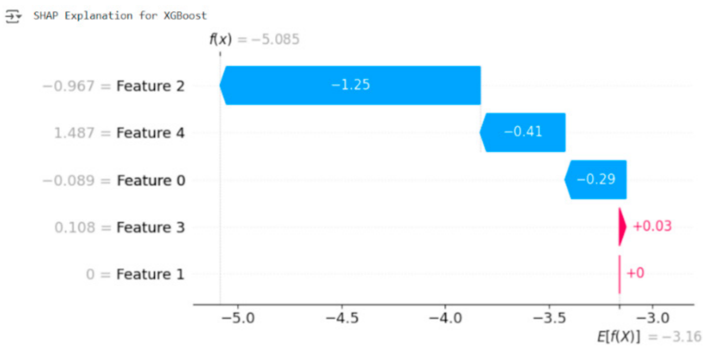


Fig. 3. SHAP explanation for XGBoost model

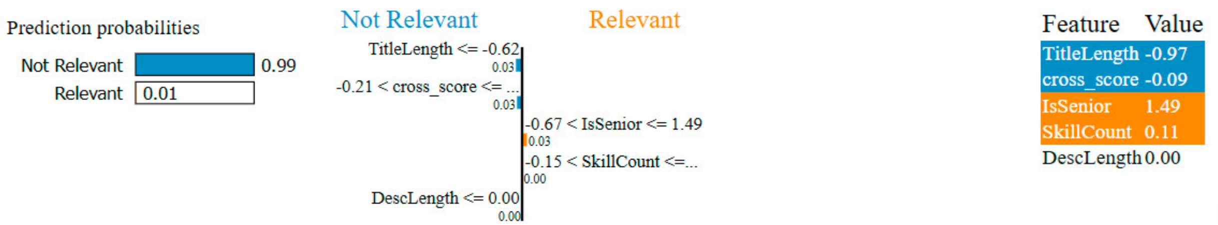
As shown in Figure 3, SHAP values reveal that features such as “SkillCount” and “IsSenior" played significant roles in influencing classification outcomes. As XGBoost natively supports SHAP’s TreeExplainer, explanations were precise and consistent.

4.4.2. LIME analysis

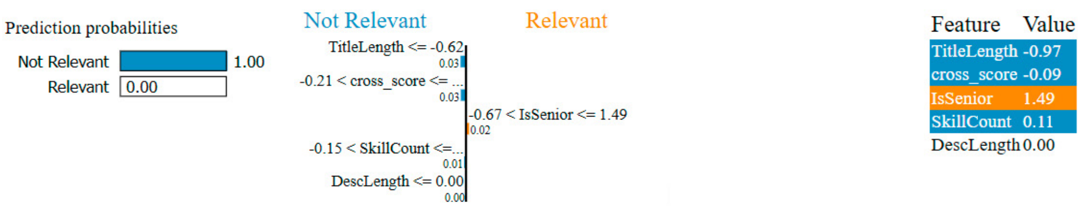
The LIME visualizations provide local interpretability for each model's prediction of whether a given job posting is relevant to a target profile like "data scientist". The explanation breaks down how each feature's value contributed to the predicted class either “Relevant” or “Not Relevant.”

Figures 4 depicts LIME-based local explanations for the three employed ensemble models. LIME was successfully applied across all classifiers and generated interpretable local feature attributions for individual recommendations. This demonstrates LIME’s generalizability across model types and its utility in explaining recommendations even in non-gradient models.

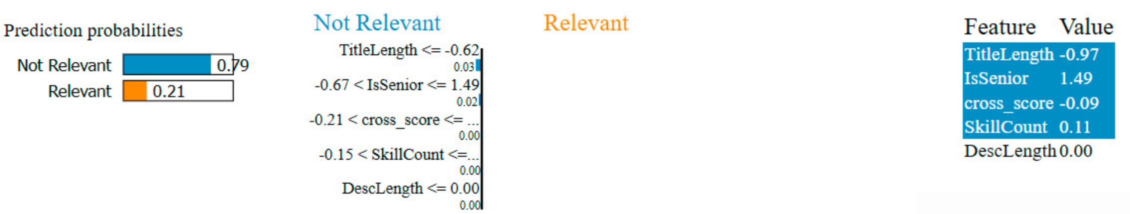
Considering XGBoost model, as shown in Figure 4 (a), it strongly penalizes short job titles and weak semantic similarity. However, it assigns non-zero weight to "IsSenior" and "SkillCount" features, indicating nuanced decision-making. This suggests that while the semantic signal wasn’t convincing, the feature-level match was borderline. Moreover, XGBoost maintains a slightly better balance between semantics and structure compared to Random Forest. On the other hand, as shown in Figure 4 (b), Random Forest makes a definitive prediction that the job is not relevant. It highly penalizes short job titles, low semantic similarity, and minimal skill matches. The model tends to heavily favor structured feature thresholds indicating its sensitivity to numerical patterns and offers less flexibility in borderline cases than AdaBoost.



(a) XGBoost model



(b) Random Forest model



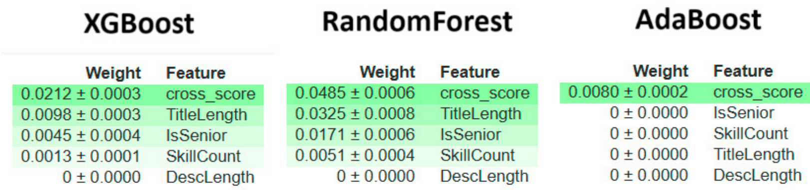
(c) AdaBoost model

Fig. 4 LIME explanation for ensemble models

The AdaBoost model as shown in Figure 4 (c), penalizes short job titles and low cross-encoder similarity scores. Although “IsSenior” was within a moderate range, the semantic signal "cross_score" and shallow “TitleLength” contributed most toward the "Not Relevant" prediction. Accordingly, AdaBoost appears conservative, requiring strong positive signals to flip its prediction.

4.4.3. ELI5 analysis

By simplifying the rationale behind model decisions into layman’s terms, ELI5 increases trust and usability for non-technical users. This aligns with recent literature advocating for multi-layered explainability in recommendation systems and supports the system’s adoption in high-stakes environments like hiring.



XGBoost

Weight	Feature
0.0212 ± 0.0003	cross_score
0.0098 ± 0.0003	TitleLength
0.0045 ± 0.0004	IsSenior
0.0013 ± 0.0001	SkillCount
0 ± 0.0000	DescLength

RandomForest

Weight	Feature
0.0485 ± 0.0006	cross_score
0.0325 ± 0.0008	TitleLength
0.0171 ± 0.0006	IsSenior
0.0051 ± 0.0004	SkillCount
0 ± 0.0000	DescLength

AdaBoost

Weight	Feature
0.0080 ± 0.0002	cross_score
0 ± 0.0000	IsSenior
0 ± 0.0000	SkillCount
0 ± 0.0000	TitleLength
0 ± 0.0000	DescLength

Fig. 5. ELI5 explanation for XGBoost, Random Forest and AdaBoost

Figure 5 demonstrates the ELI5 explanation overlay across all models. The results demonstrate a consistent trend in which “cross_score”, the semantic similarity derived from the BERT-based Cross-Encoder emerges as the most influential feature in all models. XGBoost assigns it the highest importance (0.0212 ± 0.0003), followed by “TitleLength”, “IsSenior”, and “SkillCount” features, indicating that this model effectively integrates both semantic and structural information. Similarly, Random Forest exhibits a clear descending weight gradient from “cross_score” (0.0485 ± 0.0006) to “DescLength”, reflecting the model's capacity to balance textual and engineered features in its decision logic. In contrast, AdaBoost reveals a sharply concentrated importance solely on “cross_score” (0.0080 ± 0.0002), with zero weight assigned to all other features.

This suggests a high degree of model bias and reliance on a single semantic indicator, reinforcing prior observations of underutilization of auxiliary features. Overall, ELI5’s output confirms that ensemble classifiers prioritize semantic relevance as a dominant factor while models like XGBoost and Random Forest also leverage additional structural signals to refine recommendation decisions. The results confirm that the proposed hybrid system, which combines semantic similarity via BERT-based Cross-Encoder and ensemble classification models, significantly enhances recommendation quality.

4.5. Limitations of Explainability

Despite the notable performance if the proposed hybrid system, it faces limitations in interpreting incorrect predictions. SHAP's inconsistent compatibility across ensemble classifiers and its abstract output hinder non-technical user understanding. To address this, ELI5 was integrated for clearer explanations. In addition, incorrect recommendations primarily stem from semantic misalignment or underutilized auxiliary features, especially when Cross-Encoder scores are low despite strong structural matches. This highlights the critical role of semantic similarity. Moreover, while a BERT-based Cross-Encoder replaced previous lexicon-biased methods, its shallow architecture (MiniLM) may underperform in capturing complex domain-specific relationships. This merits further investigation into more optimized Cross-Encoders for improved semantic precision. Fundamentally, enhancing XAI robustness and developing deeper semantic modelling are crucial for improving the accuracy and trustworthiness of job recommendation systems.

4.6. Social Impact and Practical Implications of Explainable Job Recommendation Systems

The proposed explainable job recommendation system significantly enhances e-recruitment fairness and accessibility, especially for marginalized groups. By combining BERT-based understanding with XAI techniques;

namely SHAP and LIME, it delivers transparent, accurate recommendations that foster user trust and align with ethical AI. Practically, the system streamlines hiring by providing AI-driven, semantically relevant candidate ranking, reducing mismatches and boosting efficiency. Its explainability empowers HR decision makers to audit decisions, mitigate bias, and ensure compliance, eventually leading to improved hiring outcomes and organizational performance through reliable, transparent support.

5. Conclusion and Future Work

This study presents a hybrid and explainable job recommendation system that integrates BERT summarization, Cross-Encoder semantic similarity, ensemble classifiers, and multi-layered explainability techniques. Each layer of the model contributes uniquely. BERT provides deep contextual alignment, classifiers capture structured relevance patterns, and explainability modules offer transparent justifications. Together, they ensure that recommendations are accurate, relevant, and trustworthy. By combining semantic signals with structured features, the system significantly improves recommendation performance and interpretability. Among classifiers, XGBoost showed superior performance (93.4% accuracy, RMSE = 0.237), while the explainability layer via SHAP, LIME, and ELI5 enhanced trustworthiness and model transparency. In conclusion, the system successfully overcomes the limitations of traditional lexical matching approaches such as TF-IDF and Cosine similarity by employing a Cross-Encoder based semantic matching strategy.

Future work will explore extending the proposed system by incorporating knowledge graphs to further enrich job-user matching through representing entities such as job roles, required skills, certifications, and industries in a structured, interconnected format. This enables reasoning over indirect relationships and improves recommendations for emerging roles or cross-domain profiles. Moreover, integrating graph-based embeddings and ontology-aware similarity measures can be considered to enhance contextualization, reduce cold-start issues, and support multilingual and cross-domain recruitment scenarios.

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